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adoption of benchmarks and to gain awareness of them. So we have a write-up that's available in archives. So if you're interested in reading about some of the challenge and recommendations from the workshop, that's also available. Okay, so that brings us to our benchmarking vision, which came out of that workshop. And we want to have this centralized resource for biologically relevant benchmarks, data sets, and tasks alongside dynamic, interpretable, up-to-date results. And we can only really do this with community contribution. So we really want to make it easy for people to contribute these benchmarking assets. And we're thinking about the community in three different ways. So one, we're spinning up domain-specific working groups to get experts together to help define the assets and come up with more complex tasks that are actually useful for the experimentalists. And then individual community contribution. So if you have a new model and you run our benchmarking package and you want to contribute your model and have other people be able to benchmark it easily, you can do that as well. And then we're also working with different partners since there are so many great benchmarking efforts that are already available. So we're working closely with them to make sure that we're just kind of supercharging what we're offering and not duplicating efforts. So the first thing that we developed is the CZ benchmarking package, and it provides a toolkit for developers to run containerized models that are also available on the Virtual Cell platform. And then we have curated evaluation data sets available that can be run on biologically relevant tasks and pre-computed performance metrics. So we are leveraging these benchmarks for standardized evaluation and comparison of models for biological applications. The package is right now available on GitHub. So there's a QR code if you want to check it out. It's under active development, but we would love to get feedback on it. So please check it out. It also facilitates interoperability with our virtual cell platform. So we have the models available on the platform and the data sets available on the platform with additional assets around model cards and data cards that describe exactly how curation was done and what's in the data set. So you can use this CLI command to just pull in these assets and then run on predefined combinations of tasks with which models should be benchmarked and on which evaluation data set. And then we also have the metrics too. And then the CZ benchmarks will serve the results on the platform. So we've run that and we have the results available on the platform. So you can view it in a table form and kind of configure exactly what you want to see to answer the question you're asking. So you can group by data set, you can filter for specific tissue types, whatever you're interested in, or sort by metrics of most interest. And then we also have a heat map available for easy visualization. So where are we now? Right now we've been focusing on single-cell transcriptomics and we have three widely used tasks available, 11 models benchmarked and 10 metrics across the three tasks. It's up on our platform right now. Again, I put a QR code if you want to check it out. And you can also link to the package or documentation from there. Community contributive benchmarks are coming. So earlier this year, we hosted a Jamboree to bring together ML researchers and computational biologists specializing in single-cell

transcriptomics. And the idea was to establish benchmarks for single-cell models trained on observational data. And it served as a kickoff for our first working group, which is working on defining assets to be added to what we're already offering. And that should be added within the next couple of months. And the idea is that we'll kind of seed with like traditionally used tasks, and then the community can continue contributing and building out what we're offering. We also have this billion cells project, which is launched to generate open data to make available for training models, which is super exciting. And we're going to leverage this project to have some held out data sets available so that we can have them for evaluation to kind of try and prevent data leakage. And then eventually, after a certain amount of time, we'll make that available as well. And then we'll be expanding to other domains. So right now we're starting with single-cell transcriptomics, but if we want to get to the virtual cell, we'll need to go across different domains and modalities. So we'll be expanding our benchmarking package to offer variant effect prediction, spatial transcriptomics, and dynamic imaging. And so we'll scale impact through interoperability and access to high-quality cross-domain biologically focused assets. And so we would love for you to contribute, and there's a few ways you can do that. You can work with us closely right now to contribute assets. We're like working on building the infrastructure to make it easy for anyone to contribute. But in this phase where we're still in beta, we're kind of working closely with community members to contribute different assets. We also are asking for feedback on our package because we want to make it as easy to use and useful for the community so we can get broad adoption. So you can either come find me after this session. We also have a booth downstairs. It's booth 13 and there are other CZI employees there or you can send me an email and we can set up time to chat. Thank you. Thanks Louise for the nice presentation and all the work you're doing for the community and also for supporting this. So I think yes, if we want to stay on time, we're going to leave the questions for the next one. So thanks actually to Gustavo. We have managed at the beginning of the day. So in the simplest form, we want people to be able to say which model performs best for this task, but we also recognize that benchmarking is really critical for advancing AI and biology. And it can do this through these three essential functions. So one, of course, it builds trust through rigorous model validation, but it's also able to accelerate development in areas that are important to biologists. So it can create this flywheel effect where we can see where models aren't performing as well, data can be generated in those areas, and then models can increase performance. And it's also just a natural collaboration point for biologists and developers so that experimentalists can help define tasks that are interesting to benchmark on and then also define metrics that are biologically relevant. So last year, towards the end of the year, we hosted a workshop with external experts to come across different domains to talk about the current state of benchmarking across domains and kind of see where the challenges are and potential solutions that we can come up with. And there are a lot of learnings. I'm going to just share a couple here. But even though there were unique

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